MASTER'S IN BUSINESS ANALYTICS · DEFENSE JULY 2026

# Progressive Equipment Recognition with Computer Vision


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OUTLINE

# What I will cover

01 Context and problem

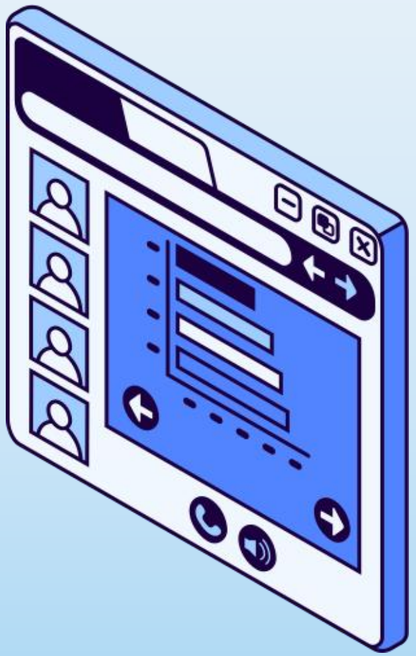
02 Solution and method

03 V1 — RAG baseline

04 V2 — EquipFace

05 V3 — Segmentation

06 Demo and perspectives



# Host organization

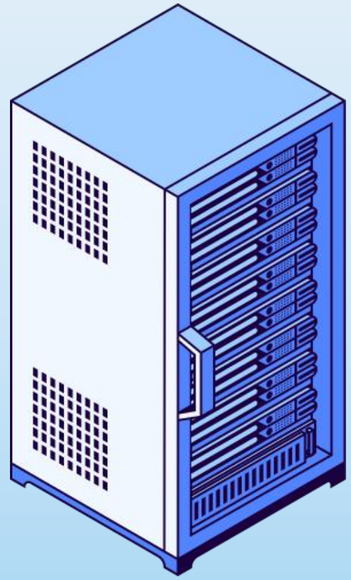
SFM Technologies Tunisie · engineering & IT services

**1995**
Founded in Tunis

**4**
Activity poles

**5**
Continents served

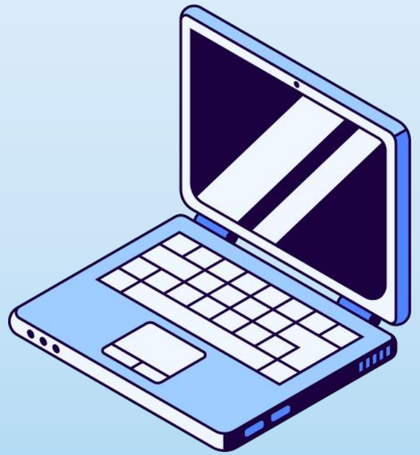
**AI**
Host team: AI & Mobile



# Today: four manual steps

Current workflow

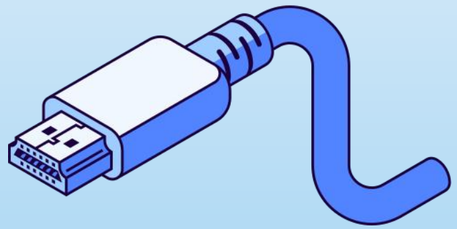
1. Inspect or photograph on site
2. Identify by eye or by label
3. Search inventory by reference
4. Update the record by hand



# What breaks

- Slow at audit scale
- Lookup errors create inventory mismatches
- Accuracy depends on the operator
- Worn label = no ground truth

> No image search exists — a photo retrieves nothing.



# The problem

*Identify a device from one photo — and retrieve its specifications.*

| HARDWARE | DATA | TIME | CATALOG | TEAM |
| --- | --- | --- | --- | --- |
| CPU only | No labeled set | 4.5 months | 5 categories | 1 developer |

# Three versions, one core

OpenCLIP · FastAPI · Streamlit

V1 · BASELINE

## RAG retrieval

Photo matched to catalog text

No enrollment · fastest

V2 · EQUIPFACE

## Few-shot match

Photo matched to photo

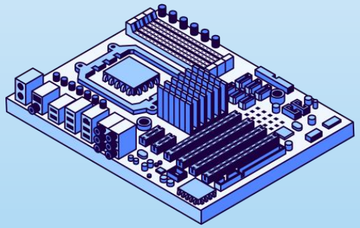
Enrollment · precise, open-set

V3 · SAM2 + EQUIPFACE

## Segment, then match

Background removed first

Enrollment · noise-robust



# Scrum roles and backlog

### Product Owner

Company supervisor

### Scrum Master

Academic advisor

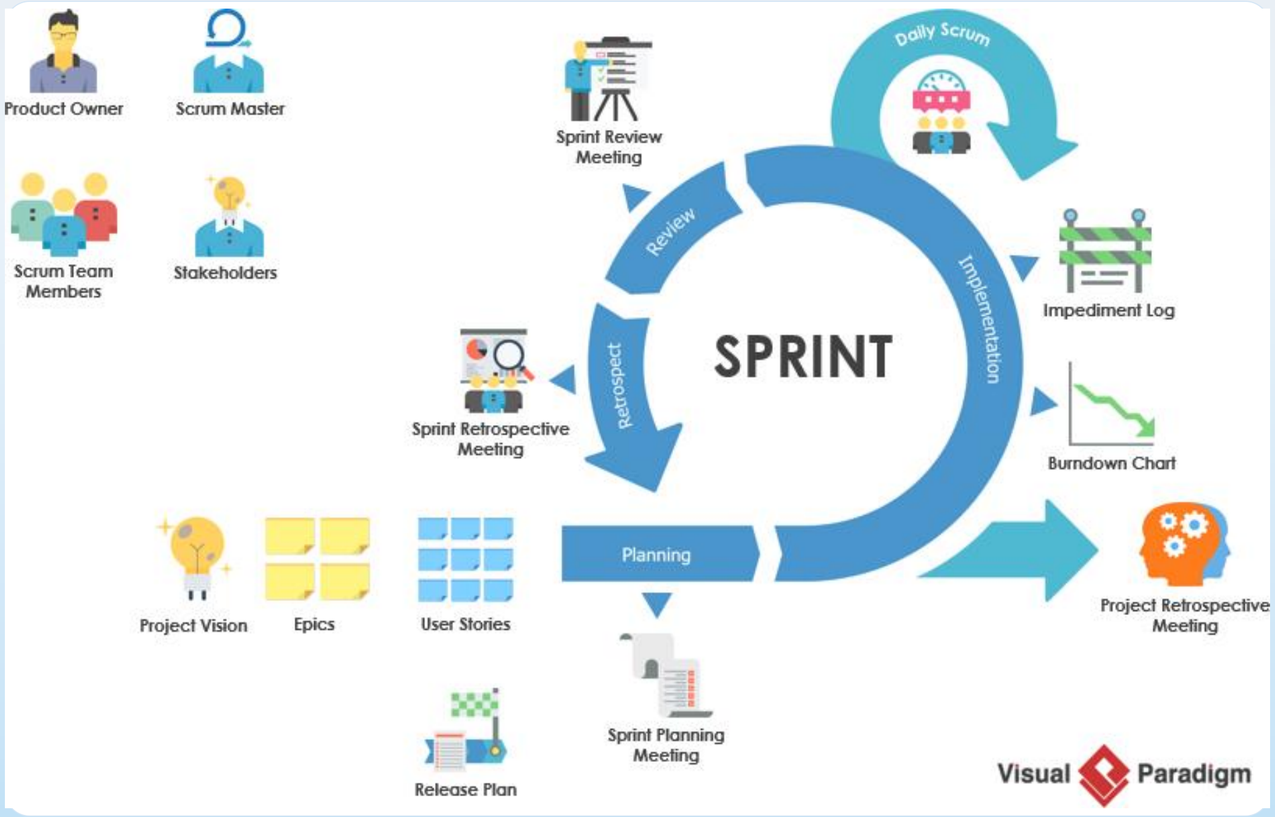
### Development team

The intern

PRODUCT BACKLOG

## 12 user stories

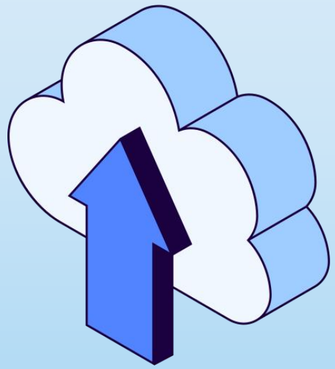
MoSCoW: 6 must · 5 should · 1 could



# Six sprints

16 February – 30 June 2026

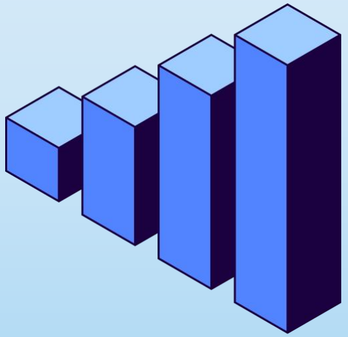
| | | |
|---|---|---|
| **Sprint 0** | Research | 16 Feb – 10 Mar |
| **Sprint 1** | V1 · RAG pipeline | 11 Mar – 1 Apr |
| **Sprint 2** | V2 · EquipFace + registry | 2 Apr – 23 Apr |
| **Sprint 3** | V3 · SAM2 segmentation | 24 Apr – 16 May |
| **Sprint 4** | Orchestrator, API, clients | 17 May – 8 Jun |
| **Sprint 5** | Benchmarking and writing | 9 Jun – 30 Jun |



# How each version is judged

Five indicators · same test set for V1, V2, V3

| | |
|---|---|
| **Top-1 accuracy** | Correct answer ranked first |
| **Top-3 / Top-5** | Ranking problem or retrieval problem |
| **MRR@5** | Typical rank of the right answer |
| **Latency** | Time per image on CPU |
| **Confidence gap** | Right answers vs wrong answers |



# V1 — semantic baseline

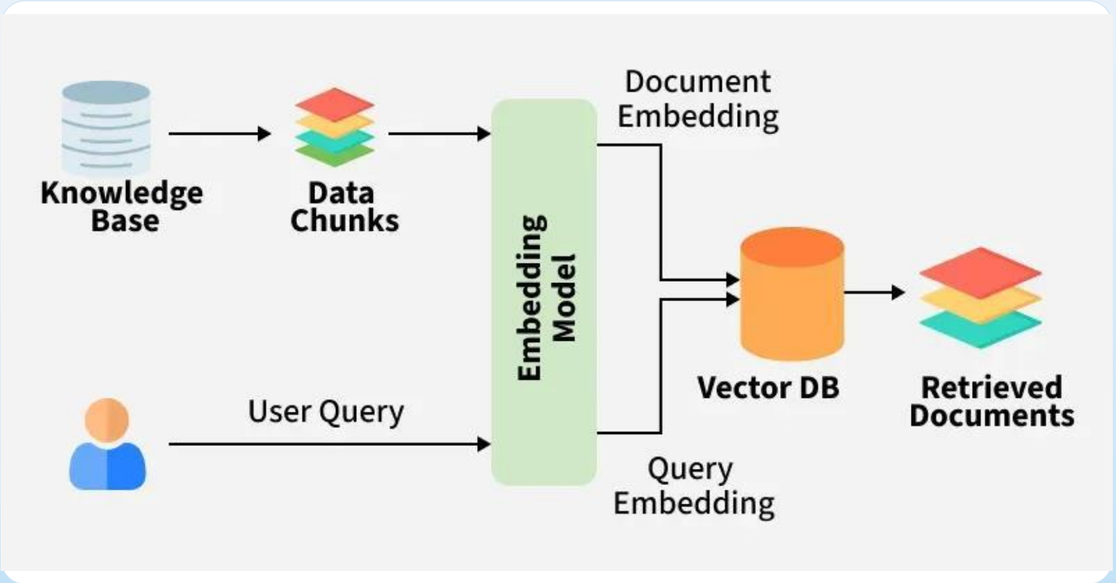
*Four pretrained components · nothing trained · CPU*

**01 DETECT** — **YOLO-World**
Categories typed as text

**02 EMBED** — **OpenCLIP**
512-dimension vector per crop

**03 RETRIEVE** — **ChromaDB**
Cosine similarity over the catalog

**04 REPORT** — **Gemini 2.5 Flash**
Structured JSON output



# V1 — results

500+ images · 79 items · CPU only

**26.2%**

**Top-1**

1 query in 4 ranked first

**41.9%**

**Top-3**

Right item in first three

**52.9%**

**Top-5**

Retrieved more often than not

**0.357**

MRR@5 — around rank 3

**116 ms**

Latency per image, CPU

**86 / 75%**

Confidence right vs wrong

# Found — but not ranked first

TOP-1 · RANKED FIRST

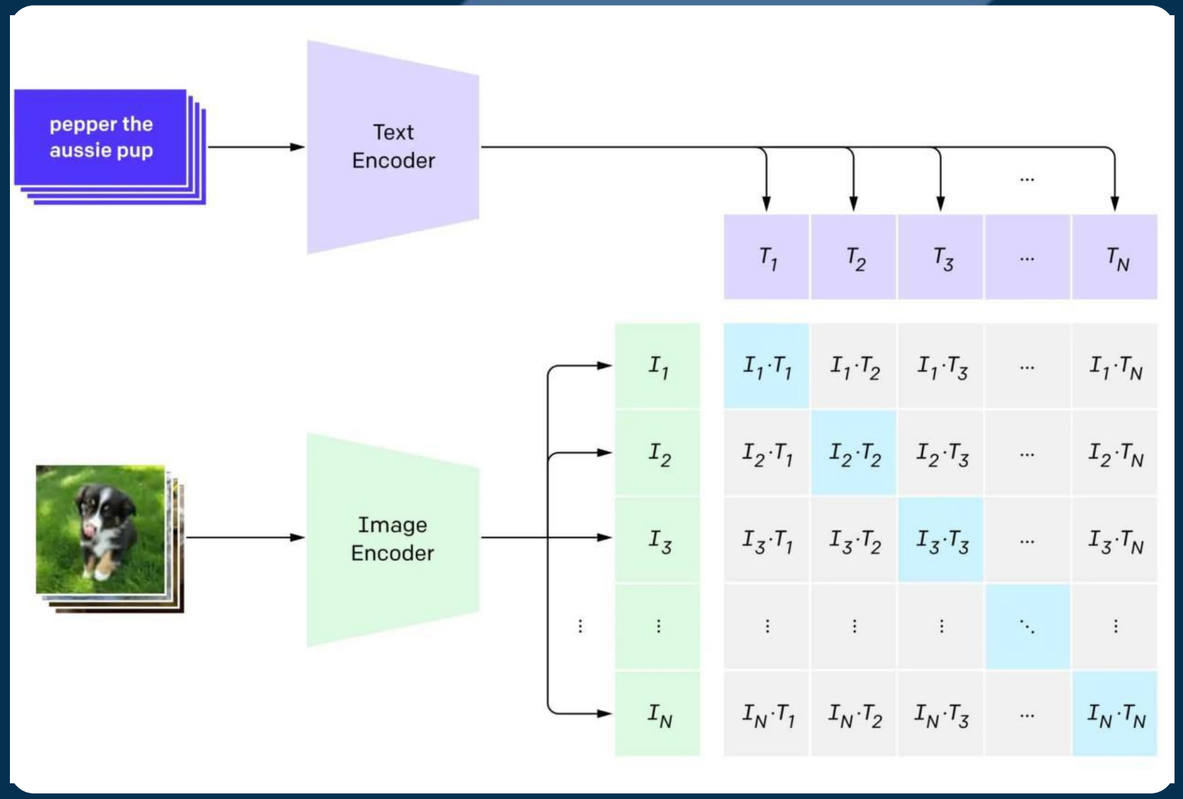
26.2%

TOP-5 · RETRIEVED AT ALL

52.9%

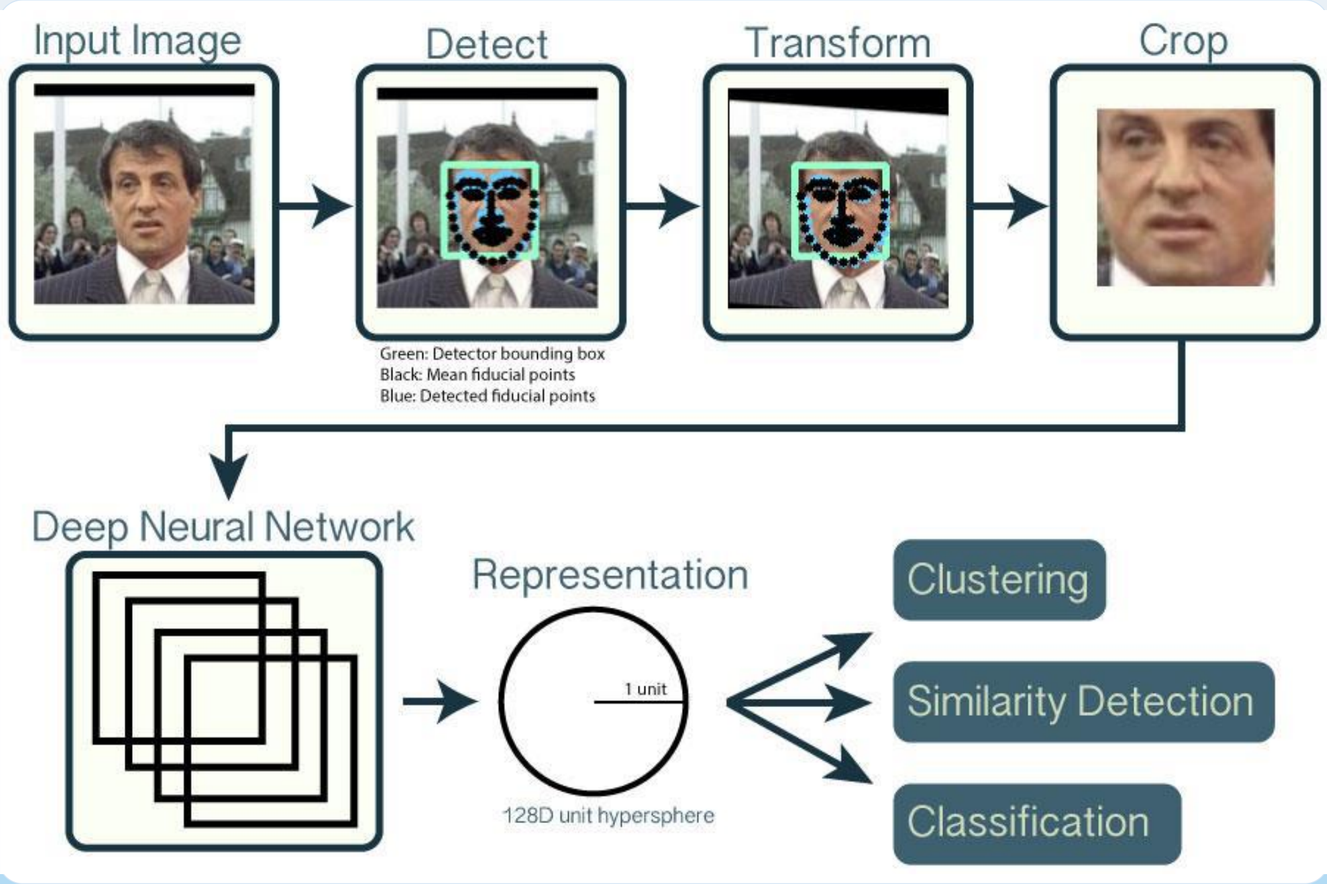
Image → text pays a penalty

Sprint 2: compare image → image



# Borrowed from face recognition

*The same recipe, applied to equipment*

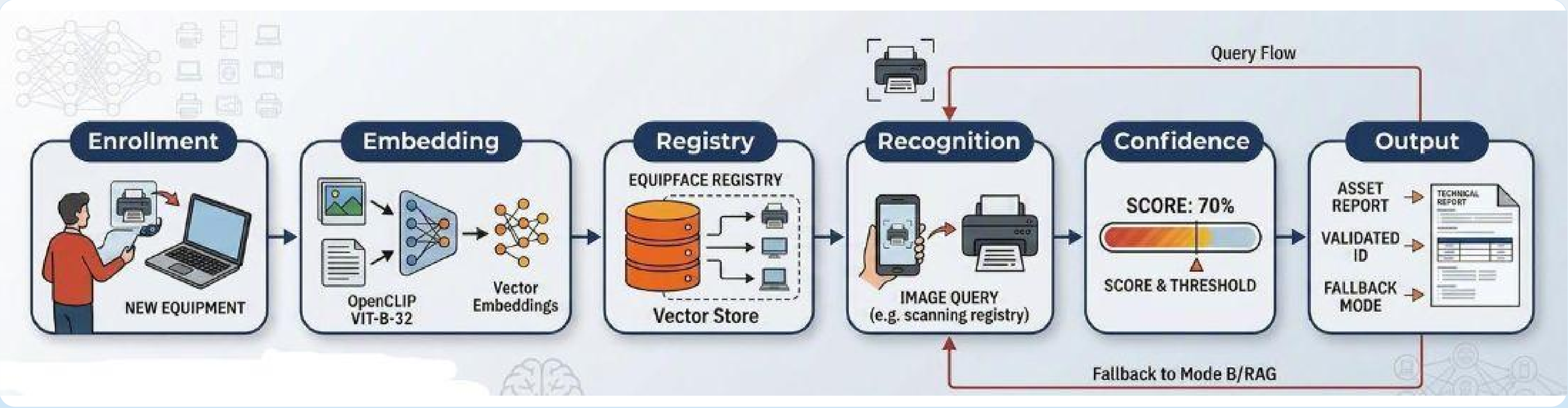



- Detect → crop → embed
- Identity = nearest reference
- New model added without retraining

> A face is a person. A chassis is a model.

# V2 — EquipFace architecture

*Enroll · embed · register · match*




**FROZEN ENCODER**

*OpenCLIP never fine-tuned*

**OPEN-SET ENROLLMENT**

*New device enrolled in seconds*

**MAX SIMILARITY**

*Best reference decides identity*

# The enrolled registry

Built from scratch · 79 → 248 models

| EQUIPMENT TYPE | MODELS | REFERENCE IMAGES |
|---|---|---|
| Printer | 87 | 348 |
| Desktop computer | 61 | 244 |
| Laptop | 41 | 154 |
| Refrigerator | 38 | 152 |
| Air conditioner | 21 | 105 |
| **Total** | **248** | **1,003** |

Views per model: front · ports · logo · panel · three-quarter

# V2 against the baseline

Same test set · same metrics

| METRIC | V1 · RAG | V2 · EQUIPFACE |
|---|---|---|
| Top-1 | 26.2% | **21.9%** |
| Top-3 | 41.9% | **34.5%** |
| Top-5 | 52.9% | **46.9%** |
| MRR@5 | 0.357 | **0.305** |
| Latency | 98 ms | **202 ms** |
| Confidence gap | 11 pts | **8 pts** |

The aggregate hides the per-category story.

# The background competes




## FIVE RECURRING FAILURES

- Identical chassis
- Reflective finishes
- Partial occlusion
- Low resolution
- Unseen viewing angles

> Clutter amplifies all five.

CHAPTER 5 · V3

# Segment, then recognize

One new stage · recognition logic unchanged

DETECT

**YOLO-World box**

NEW · SEGMENT

**SAM2 mask**

Background removed

RECOGNIZE

**EquipFace**

REPORT

**Gemini**

### WHY SAM2

Box as prompt · any category · CPU

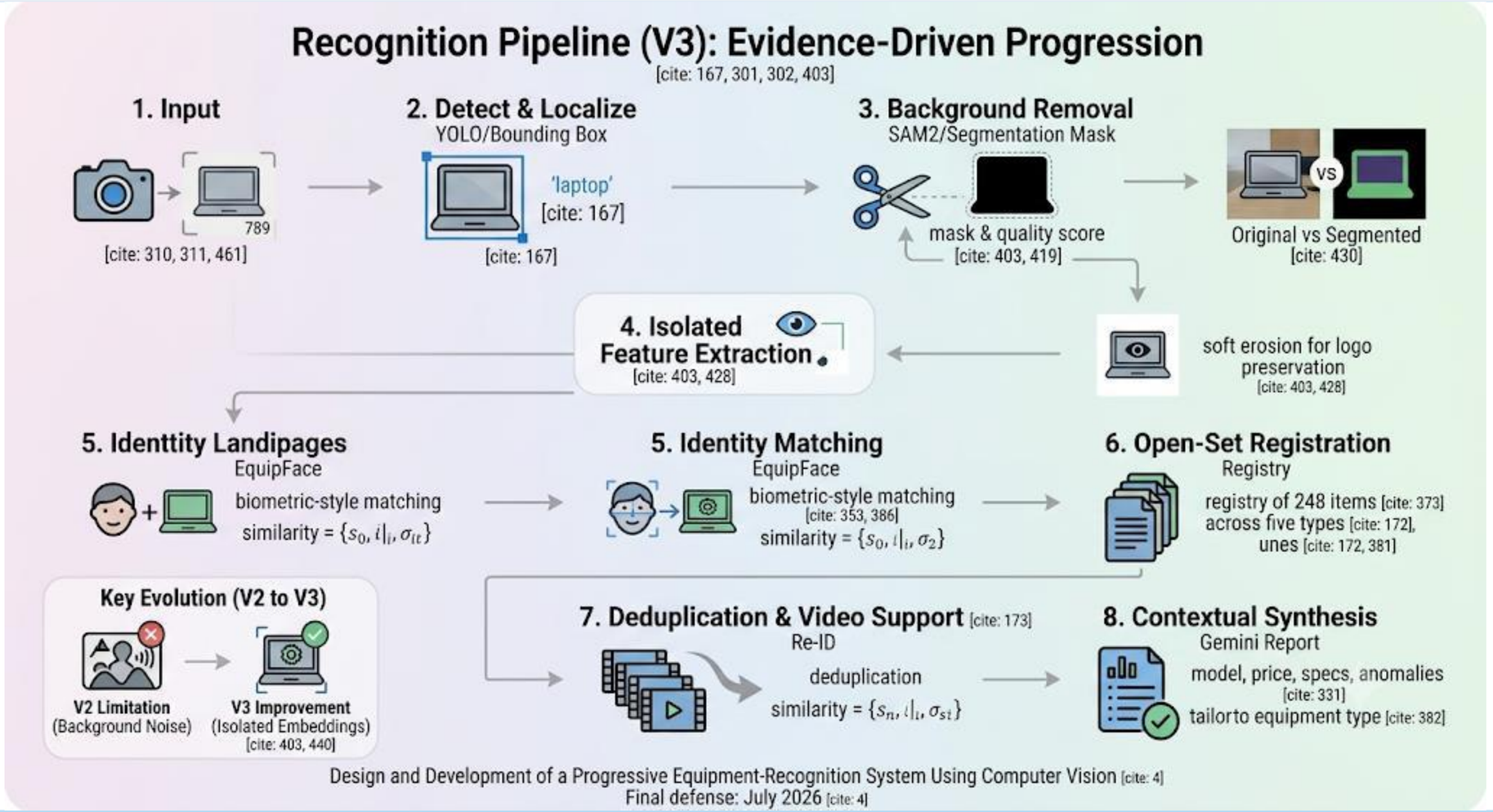
### QUALITY CHECK

Masks under 30% or over 97% rejected

### COST

2.5–5 s per image on CPU

# The full pipeline



# Built to degrade, not fail

SOFT MASKING

## Keeps the logo

- White mask erased backlit logos
- Mask eroded inward by 3 px
- +10 ms per mask

FALLBACK CHAIN

## Never crashes

- SAM2 error → V2
- Bad mask → plain crop
- Every fallback shown in the UI



# One result per device

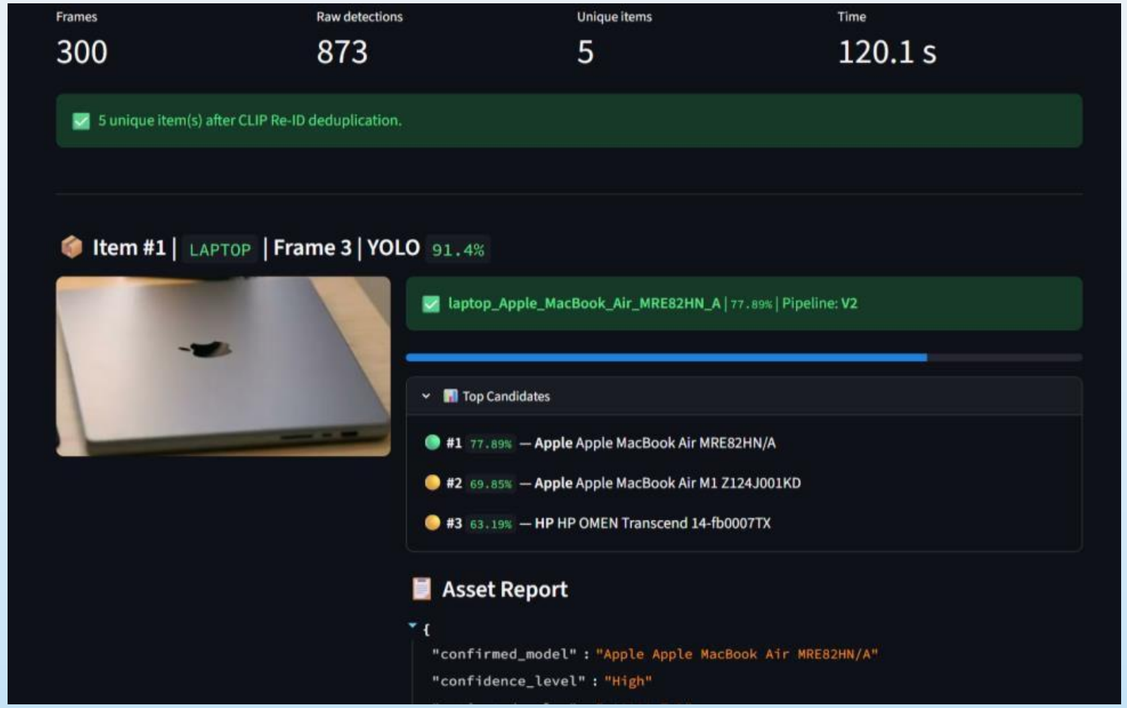
Re-ID deduplication · similarity > 0.88 = repeat

**300**
Frames

**873**
Raw detections

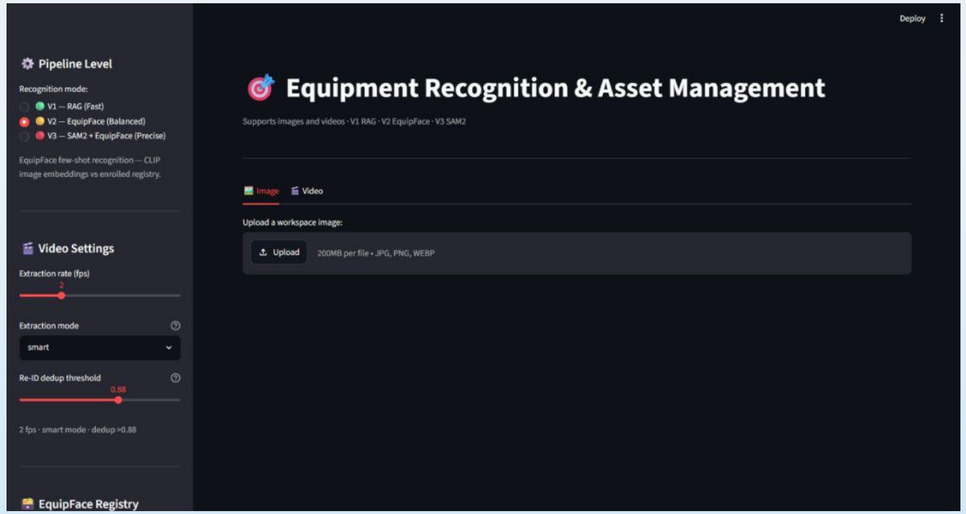
**5**
Unique items

**120 s**
End to end, CPU



# The application

Streamlit demo · FastAPI · Flutter client




Pipeline selector · live registry count




Crop · confidence · top-3 candidates

# Reports that know the type

*One JSON schema per category*

### TAILORED SCHEMA

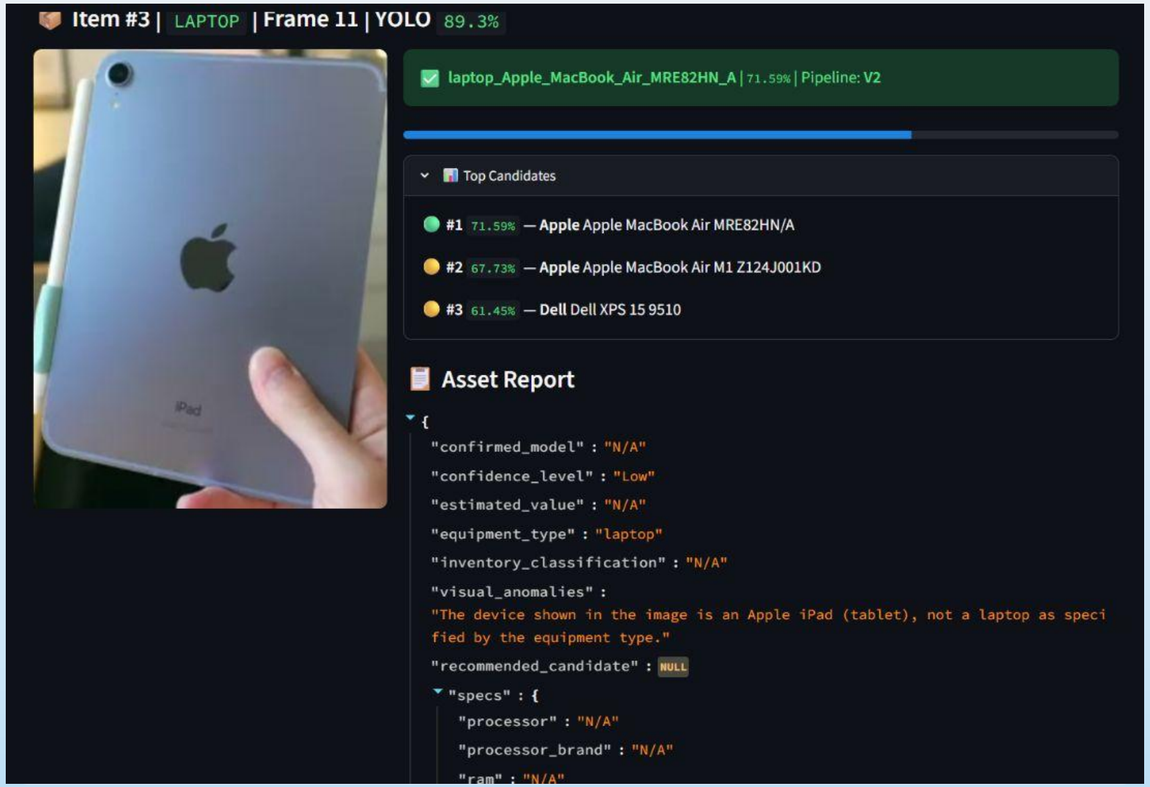
Laptop: CPU, RAM, display. Fridge: capacity, energy.

### CONFIDENCE STATED

Declines to confirm on weak evidence

### ANOMALY FLAGGED

Here: an iPad, not a laptop



SYNTHESIS

# Measured, then improved

*Every version answers a measured limitation*

| PROPERTY | V1 · RAG | V2 · EQUIPFACE | V3 · SAM2 + EQUIPFACE |
|---|---|---|---|
| Query | Image → text | Image → image | Clean image → image |
| Background | Kept | Kept | Masked |
| Latency | ~120 ms | ~200 ms | ~2.5–5 s |
| Fallback | None | None | Falls back to V2 |
| Limit | Cross-modal gap | Background noise | SAM2 latency on CPU |

# Limitations

> Bounded by data, not by models.

### Registry dependency

Not enrolled = not recognized

### Reference depth

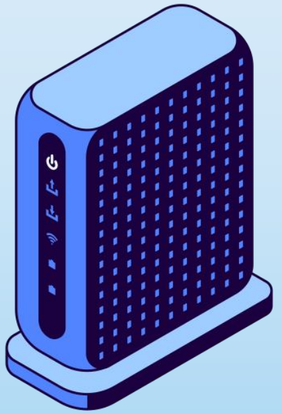
One photo is one point; ten is a cloud

### Dataset diversity

Consumer and office equipment only

### Evaluation scope

Closed-set · one site · one CPU



# What comes next

No architectural change required

### Registry enrichment

More brands and views

### Automatic enrollment

Triggered by procurement

### OCR second signal

Read labels and asset tags

### Domain fine-tuning

Widen the confidence margin

### Cloud deployment

GPU removes the latency limit

### Tracking & lifecycle

Counting and richer reporting

CONCLUSION

# Three contributions

01

## EquipFace registry

248 models, no retraining

02

## Safe segmentation

Soft masking and fallbacks

03

## Evidence-driven method

Each version earned by measurement

# Thank you — questions welcome

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